

NETWORK TRAFFIC ANALYSIS REPORT

Suspicious DNS, HTTP, and TCP Activity

KifaruCart Global Ltd.

Table 1. Document Control

Attribute	Details
Organization	KifaruCart Global Ltd.
Document Title	Forensic Network Traffic Analysis Report (Incident Ref: INC-2026-0822)
Incident Date	August, 2026
Prepared By	Lead Incident Response Analyst
Affected Endpoint	WAREHOUSE-LT-07 (10.10.20.47)
Target Audience	Security Operations Center (SOC), IT Infrastructure, CISO
Incident Status	Active Containment & Triage Phase

TABLE OF CONTENTS

<u>EXECUTIVE SUMMARY</u>	3
<u>1. INCIDENT CONTEXT AND NETWORK ENVIRONMENT</u>	3
1.1 Business and Endpoint Context	3
1.2 High-Level Traffic Flow	3
1.3 Network Asset Map	4
<u>2. DETAILED LOG ANALYSIS AND PROTOCOL DEEP-DIVE</u>	4
2.1 Domain Name System (DNS) Analysis	4
2.2 Hypertext Transfer Protocol (HTTP) Analysis	5
2.3 Transmission Control Protocol (TCP) Analysis	5
<u>3. COMBINED CHRONOLOGICAL ATTACK TIMELINE</u>	6
<u>4. KEY FINDINGS AND ATTACK CHAIN ANALYSIS</u>	7
4.1 Suspected Root Cause and Attack Vector	7
<u>5. IMMEDIATE INCIDENT RESPONSE ACTIONS</u>	7
<u>6. STRATEGIC LONG-TERM REMEDIATION</u>	8
<u>7. CONCLUSION</u>	8
<u>APPENDIX A: PROTOCOL REFERENCE GUIDE</u>	9

LIST OF TABLES

<u>Table 1. Document Control</u>	1
<u>Table 2. Network Asset Map</u>	4
<u>Table 3. DNS Log Capture</u>	4
<u>Table 4. HTTP Log Capture</u>	5
<u>Table 5. TCP Connection Summary</u>	6
<u>Table 6. Combined Chronological Attack Timeline</u>	6
<u>Table 7. Key Findings and Attack Chain Analysis</u>	7
<u>Table 8. Immediate Incident Response Actions</u>	7
<u>Table 9. Strategic Long-Term Remediation</u>	8
<u>Table 10. Protocol Reference Guide</u>	9

LIST OF FIGURES

<u>Figure 1. High-Level Network Architecture and Traffic Flow</u>	3
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EXECUTIVE SUMMARY

On August 22, 2026, the IT team detected abnormal outbound network traffic originating from WAREHOUSE-LT-07 (10.10.20.47), a Windows 11 workstation assigned to Warehouse Operations. Initial log analysis revealed anomalous high-frequency Domain Name System (DNS) queries to unverified domains, outbound Hypertext Transfer Protocol (HTTP) sessions to suspicious external IP addresses, and internal Transmission Control Protocol (TCP) connection attempts targeting core business servers.

The observed behaviors indicate endpoint compromise, initial command-and-control (C2) beaconing, malicious payload retrieval, potential staging for data exfiltration, and internal network reconnaissance consistent with lateral movement preparation.

Immediate network isolation of WAREHOUSE-LT-07 is mandated, alongside firewall blocklists for external IP addresses 203.0.113.45 and 198.51.100.77. Short-term and long-term remediation steps focus on host forensic triage, user credential resets, DNS filtering, centralized monitoring, and network segmentation.

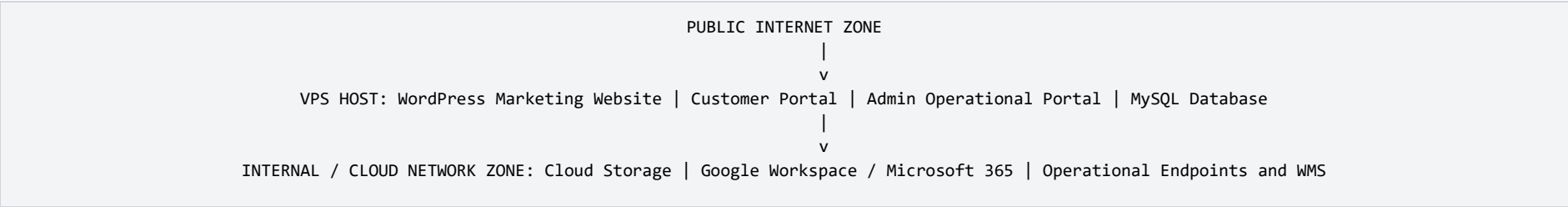
1. INCIDENT CONTEXT AND NETWORK ENVIRONMENT

1.1 Business and Endpoint Context

Kifarucart Global Ltd. operates an e-commerce and logistics platform connecting Kenyan buyers with U.S. merchants. The affected endpoint, WAREHOUSE-LT-07, resides within the Warehouse Operations VLAN (10.10.20.0/24) and is primarily used by operational staff to handle package intake and shipment tracking.

1.2 High-Level Traffic Flow

Figure 1. High-Level Network Architecture and Traffic Flow



1.3 Network Asset Map

Table 2. Network Asset Map

Host / Asset Name	IP Address	Infrastructure Role & Context
WAREHOUSE-LT-07	10.10.20.47	Suspected compromised workstation (Warehouse VLAN).
Internal DNS Server	10.10.10.10	Primary internal domain resolution service.
Customer Portal Server	10.10.30.15	External e-commerce customer portal host.
Admin Portal Server	10.10.30.20	Internal operations & admin dashboard server.
MySQL Database Server	10.10.30.25	Production database storing customer PII & order records.

Host / Asset Name	IP Address	Infrastructure Role & Context
Firewall Gateway	10.10.20.1	Default gateway routing Warehouse VLAN traffic.
Suspicious External Host 1	203.0.113.45	Unverified external C2 destination.
Suspicious External Host 2	198.51.100.77	Unverified external staging destination.

2. DETAILED LOG ANALYSIS AND PROTOCOL DEEP-DIVE

2.1 Domain Name System (DNS) Analysis

DNS translates human-readable domain names into IP addresses required for network routing. Standard client operation involves sporadic lookups to trusted internal resources or well-known external domains. In this case, the affected endpoint generated repeated DNS requests to unfamiliar and pseudo-random domains within a narrow time window.

Table 3. DNS Log Capture

Time	Source IP	Source Host	Query Domain	Type	Response	Security Classification
09:14:02	10.10.20.47	WAREHOUSE-LT-07	updates.microsoft.com	A	20.54.36.229	Normal System Traffic
09:15:18	10.10.20.47	WAREHOUSE-LT-07	kifarucart.local	A	10.10.30.15	Normal Internal Traffic
09:17:43	10.10.20.47	WAREHOUSE-LT-07	xj29sd-checkin.example	A	203.0.113.45	Suspicious C2 Check-in
09:17:48	10.10.20.47	WAREHOUSE-LT-07	xj29sd-checkin.example	A	203.0.113.45	Repeated Beaconing
09:18:11	10.10.20.47	WAREHOUSE-LT-07	cdn-update-service.example	A	198.51.100.77	Suspicious Masquerading
09:18:29	10.10.20.47	WAREHOUSE-LT-07	kifaru-admin.local	A	10.10.30.20	Internal Service Lookup
09:19:03	10.10.20.47	WAREHOUSE-LT-07	a91k2l3j4.example	TXT	NXDOMAIN	Anomalous DGA Pattern
09:19:09	10.10.20.47	WAREHOUSE-LT-07	b82m4x9p1.example	TXT	NXDOMAIN	Anomalous DGA Pattern
09:19:16	10.10.20.47	WAREHOUSE-LT-07	c73q8z6r5.example	TXT	NXDOMAIN	Anomalous DGA Pattern

The DNS activity is abnormal because the endpoint queried unfamiliar destinations, generated repeated lookup bursts, and issued TXT queries to pseudo-random domains that returned NXDOMAIN. This pattern suggests malware check-in activity and possible Domain Generation Algorithm fallback behavior.

2.2 Hypertext Transfer Protocol (HTTP) Analysis

HTTP governs web application communication. GET requests retrieve remote assets, while POST requests send client data to servers. User-Agent strings normally identify client applications, but they can be spoofed by malware to blend into normal traffic.

Table 4. HTTP Log Capture

Time	Source IP	Dest IP	Method	URI	Status	Bytes Out	User-Agent	Security Analysis
09:18:14	10.10.20.47	198.51.100.77	GET	/update/check	200	512	Mozilla/5.0	Suspicious external check-in

Time	Source IP	Dest IP	Method	URI	Status	Bytes Out	User-Agent	Security Analysis
09:18:21	10.10.20.47	198.51.100.77	POST	/api/session	200	4,812	Mozilla/5.0	C2 session initialization
09:18:40	10.10.20.47	198.51.100.77	POST	/upload/logs	200	48,920	Mozilla/5.0	Potential Data Exfiltration
09:19:27	10.10.20.47	203.0.113.45	GET	/payload.bin	200	1,024	Windows-Update-Agent	Malicious Payload Stage 2
09:20:10	10.10.20.47	203.0.113.45	POST	/gate.php	200	72,440	Windows-Update-Agent	C2 Gateway Exfiltration

The HTTP activity is highly suspicious because the workstation contacted unverified external IP addresses, retrieved /payload.bin, used Windows-Update-Agent against non-Microsoft infrastructure, and transmitted unusually large outbound POST requests to /upload/logs and /gate.php.

2.3 Transmission Control Protocol (TCP) Analysis

TCP provides connection-oriented transport. A connection is initiated using a TCP SYN packet as part of the three-way handshake. In this incident, repeated SYN traffic to sensitive internal services indicates possible service discovery and lateral movement staging.

Table 5. TCP Connection Summary

Time	Source IP	Dest IP	Dest Port	Target Service	TCP Flag	Result	Security Interpretation
09:20:55	10.10.20.47	10.10.30.20	443	Admin Portal	SYN	Allowed	HTTPS reachability test
09:21:03	10.10.20.47	10.10.30.25	3306	MySQL DB	SYN	Blocked	Internal Reconnaissance
09:21:05	10.10.20.47	10.10.30.25	3306	MySQL DB	SYN	Blocked	Port Probe / Brute-force
09:21:10	10.10.20.47	10.10.30.15	80	Customer Portal	SYN	Allowed	HTTP reachability test
09:21:16	10.10.20.47	10.10.30.15	443	Customer Portal	SYN	Allowed	HTTPS reachability test
09:21:22	10.10.20.47	10.10.30.20	22	Admin (SSH)	SYN	Blocked	Internal Reconnaissance
09:21:29	10.10.20.47	10.10.30.20	3389	Admin (RDP)	SYN	Blocked	Internal Reconnaissance
09:21:34	10.10.20.47	10.10.30.25	3306	MySQL DB	SYN	Blocked	Persistent DB Probe

The TCP pattern is abnormal because a warehouse endpoint has no valid business need to initiate direct connections to database port 3306, SSH port 22, or RDP port 3389. The repeated SYN packets across the 10.10.30.0/24 subnet indicate an automated internal sweep targeting administrative services.

3. COMBINED CHRONOLOGICAL ATTACK TIMELINE

The following timeline correlates the DNS, HTTP, and TCP evidence into a single attack chain view.

Table 6. Combined Chronological Attack Timeline

Time Frame	Phase of Attack Chain	Technical Events & Evidence Observed
09:17	C2 Initialization	Device initiates DNS lookups for C2 server (xj29sd-checkin.example).
09:18	C2 Channel Setup	Connects to 198.51.100.77 over HTTP; initiates session and uploads 48.9 KB of data.
09:19	Staging & Fallback	Downloads payload.bin using a spoofed User-Agent; initiates DGA queries upon failure.

Time Frame	Phase of Attack Chain	Technical Events & Evidence Observed
09:20	Data Exfiltration	Transmits 72.4 KB of data via HTTP POST to /gate.php on 203.0.113.45.
09:21	Lateral Movement	Initiates internal port scanning against DB (3306), SSH (22), and RDP (3389).

4. KEY FINDINGS AND ATTACK CHAIN ANALYSIS

Table 7. Key Findings and Attack Chain Analysis

Finding	Severity	Evidence	Security Interpretation
C2 communication and DGA activity	High	Repeated DNS lookups to xj29sd-checkin.example and pseudo-random domains; outbound sessions to 203.0.113.45 and 198.51.100.77.	The endpoint is likely attempting to establish or maintain command-and-control connectivity.
Second-stage malware delivery and exfiltration	Critical	GET /payload.bin followed by POST /upload/logs and POST /gate.php with more than 120 KB of outbound data.	The traffic pattern suggests payload retrieval and possible staging or transfer of local data.
Internal network reconnaissance	High	SYN attempts to MySQL 3306, SSH 22, and RDP 3389 on internal servers.	The endpoint appears to be probing internal services for lateral movement opportunities.

4.1 Suspected Root Cause and Attack Vector

Initial access was likely gained via a spear-phishing email containing a malicious attachment or through a drive-by download executed by the primary user of WAREHOUSE-LT-07. The initial dropper likely executed silently, established C2 reachability, retrieved a specialized reconnaissance payload, and attempted to map internal assets for lateral movement.

5. IMMEDIATE INCIDENT RESPONSE ACTIONS

Table 8. Immediate Incident Response Actions

Priority	Targeted Action	Responsible Owner	Objective / Safety Outcome
P1	Isolate Endpoint: Sever physical/wireless connection of WAREHOUSE-LT-07.	SOC Specialist	Immediately cuts C2 channels and prevents internal spread.
P1	Block External IPs/Domains: Add 203.0.113.45 and 198.51.100.77 to firewall blocklists.	Network Engineer	Prevents other internal devices from reaching C2 nodes.
P1	Reset User Credentials: Force password reset and revoke active sessions for user.	Identity Admin	Prevents unauthorized credential use on other portals.
P1	Volatile Memory & Log Preservation: Capture RAM image and preserve local event logs.	Forensic Analyst	Preserves forensic evidence for root-cause analysis.
P2	SIEM Historical Sweep: Query proxy/firewall logs for connections to identified C2 destinations.	SOC Lead	Confirms whether other endpoints on the network are infected.
P2	Audit DB & Admin Logs: Review access logs on 10.10.30.20 and 10.10.30.25.	Database Admin	Verifies whether unauthorized database access occurred.

6. STRATEGIC LONG-TERM REMEDIATION

Table 9. Strategic Long-Term Remediation

Domain Area	Recommended Security Control	Strategic Mitigation Purpose
Endpoint Protection	Deploy Endpoint Detection & Response (EDR).	Enables behavioral detection, process kill, and automated host isolation.
DNS Security	Implement Secure DNS Filtering (e.g., Cisco Umbrella).	Automatically blocks lookups to newly registered domains and DGA patterns.
Network Architecture	Enforce Strict VLAN Micro-segmentation.	Blocks direct network paths between Warehouse VLAN and internal databases.
SIEM & Monitoring	Deploy Centralized Log Collector with automated alerts.	Triggers alerts on unexpected POST spikes or internal SYN port sweeps.
User Training	Conduct targeted phishing simulation training.	Reduces user susceptibility to initial access vector email attachments.

7. CONCLUSION

The network traffic analysis indicates that WAREHOUSE-LT-07 exhibited multiple behaviors consistent with endpoint compromise. The sequence of suspicious DNS queries, outbound HTTP payload activity, and internal TCP scanning suggests that the host progressed from C2 discovery to payload retrieval, possible data exfiltration, and internal reconnaissance.

The immediate priority is to contain the affected endpoint, preserve evidence, reset exposed credentials, block the suspicious destinations, and review related logs for signs of wider compromise. Longer-term security improvement should focus on endpoint detection and response, DNS filtering, network segmentation, centralized logging, and targeted phishing awareness training.

APPENDIX A: PROTOCOL REFERENCE GUIDE

Table 10. Protocol Reference Guide

Protocol / Term	Default Port / Transport	Purpose	Security Relevance in This Case
Domain Name System (DNS)	UDP/TCP 53	Translates human-readable hostnames into IP routing addresses.	Repeated lookups to unfamiliar and pseudo-random domains indicate possible C2 discovery and DGA fallback behavior.
Hypertext Transfer Protocol (HTTP)	TCP 80; HTTPS uses TCP 443	Carries client-server web requests such as GET and POST.	Outbound GET and POST sessions to unknown infrastructure suggest payload retrieval, beaconing, and possible data exfiltration.
Transmission Control Protocol (TCP)	Transport protocol; service port varies	Provides reliable, stateful communication through connection setup such as SYN, SYN-ACK, and ACK.	Repeated SYN attempts to MySQL, SSH, and RDP indicate probing of internal services.
Domain Generation Algorithm (DGA)	Not a protocol	Malware technique that generates many pseudo-random domains to maintain C2 reachability.	TXT lookups returning NXDOMAIN for random-looking domains suggest automated fallback behavior.